

Chapter 6

European option pricing and the Greeks

In this chapter we consider the case of a non-dividend paying underlying, with positive price process S . The basic European derivatives are presented in section 6.1, along with their parity relations. In section 6.2 we price them in the standard Black and Scholes model. Section 6.3 introduces the Greeks and their computation for the basic European derivatives in the standard Black and Scholes model. All the formulae for computing prices and Greeks are implemented in VBA in section 6.4

6.1 Basic European derivatives

Let T be a certain maturity. An *European derivative contract* is a simple T -claim and hence completely defined by its contract function $\Phi : \mathbb{R}^+ \rightarrow \mathbb{R}$.

We consider the following basic European derivative contracts:

- The (unit) *zero coupon* (ZCB), with payoff function $\Phi(s) = 1$ (figure 6.1(a)) and hence constant 1 terminal payoff. Of course, this is a trivial contingent claim, as it is not contingent at all on the terminal level of the underlying.
- Given a certain strike $K \geq 0$, the *long forward contract* (hereinafter simply forward contract) is defined by $\Phi(s) = s - K$ (figure 6.1(c)). The terminal payoff of the forward contract is therefore $F(T) = \Phi(S(T)) = S(T) - K$. The limit case $K = 0$ corresponds to the payoff of a long position in the stock (figure 6.1(b)). The corresponding *short forward contract* has payoff $K - S(T)$.
- For a contractually specified strike $K > 0$ and a cash amount $Q > 0$, the *digital (or binary) cash or nothing* (CON) *call option* is defined by

$$\Phi(s) = Q\mathbb{1}_{\{s > K\}} = \begin{cases} Q & \text{if } s > K, \\ 0 & \text{if } s \leq K \end{cases}$$

(figure 6.1(f)). Its terminal payoff is therefore $C_{\text{CON}}(T) = \Phi(S(T)) = Q\mathbb{1}_{\{S(T) > K\}}$. The corresponding *put option* has payoff function

$$\Phi(s) = Q\mathbb{1}_{\{s \leq K\}} = \begin{cases} 0 & \text{if } s > K, \\ Q & \text{if } s \leq K \end{cases}$$

(figure 6.1(g)), the terminal payoff is $P_{\text{CON}}(T) = \Phi(S(T)) = Q\mathbb{1}_{\{S(T) \leq K\}}$.

- The *digital asset or nothing* (AON) *call option* with strike $K > 0$ is defined by

$$\Phi(s) = s\mathbb{1}_{\{s > K\}} = \begin{cases} s & \text{if } s > K, \\ 0 & \text{if } s \leq K \end{cases}$$

(figure 6.1(e)), with terminal payoff $C_{\text{AON}}(T) = \Phi(S(T)) = S(T)\mathbb{1}_{\{S(T) > K\}}$.

The corresponding *put option* is defined by the payoff function

$$\Phi(s) = s\mathbb{1}_{\{s \leq K\}} = \begin{cases} 0 & \text{if } s > K, \\ s & \text{if } s \leq K \end{cases}$$

(figure 6.1(h)), and its terminal payoff is $P_{\text{AON}}(T) = \Phi(S(T)) = S(T)\mathbb{1}_{\{S(T) \leq K\}}$.

- A *plain vanilla call option* with strike $K > 0$ has payoff function

$$\Phi(s) = (s - K)^+ = \begin{cases} s - K & \text{if } s > K, \\ 0 & \text{if } s \leq K \end{cases}$$

(figure 6.1(f)); its terminal payoff is therefore $C(T) = \Phi(S(T)) = [S(T) - K]^+$.
The payoff function of the *plain vanilla put option* with the same strike is

$$\Phi(s) = (K - s)^+ = \begin{cases} 0 & \text{if } s > K, \\ K - s & \text{if } s \leq K \end{cases}$$

(figure 6.1(i)), and its terminal payoff is $P(T) = \Phi(S(T)) = [K - S(T)]^+$.

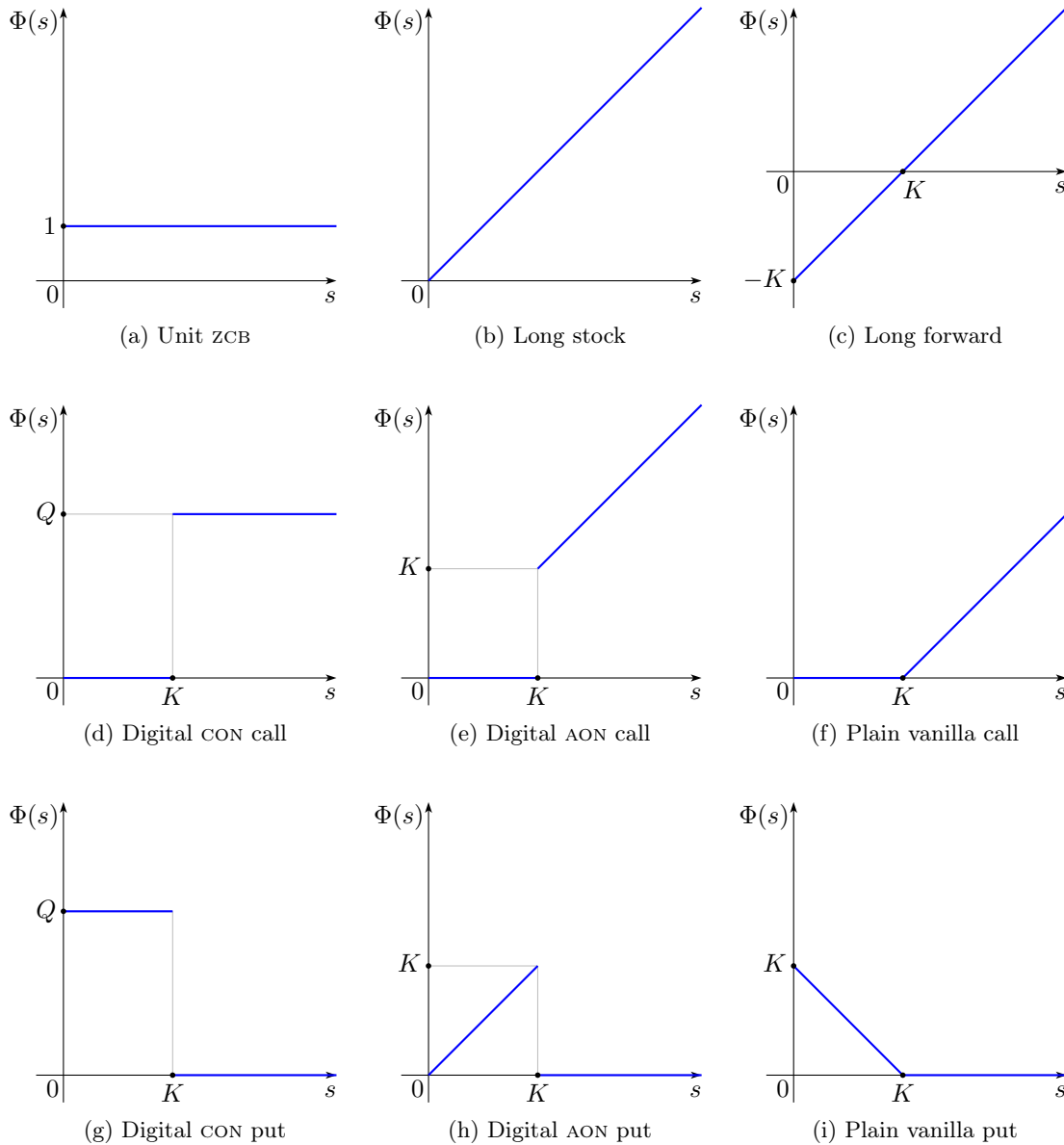


Figure 6.1. Basic European payoff functions

6.1.1 Parity relations and first pricing results

Looking at figure 6.1, it is quite evident that there are simple relationships that bind together the payoffs of the contracts in section 6.1. Let us summarise the most important of these, the so-called *parity relations*.

Theorem 6.1 (Parity relations). *If all the contracts have the same underlying S , the same maturity T , the same strike K (if relevant) and the same cash amount Q (if relevant), then:*

- (i) Cash and carry spot-forward parity. *A buy and hold portfolio composed by a long position in the stock and a short position in K unit ZCBs replicates the long forward contract with strike K ; that is*

$$F(T) = S(T) - K \ .$$

Therefore, for every $t < T$, the price at time t of the long forward contract is

$$F(t) = S(t) - Kp(t, T) \ .$$

- (ii) Put-call parity for digital CON options. *A buy and hold portfolio composed by a long digital CON call option and a long digital CON put option replicates Q unit ZCBs. That is*

$$C_{\text{CON}}(T) + P_{\text{CON}}(T) = Q \ .$$

Therefore, for every time $t < T$, the relation between the prices $C_{\text{CON}}(t)$ and $P_{\text{CON}}(t)$ at time t of the two options is

$$C_{\text{CON}}(t) + P_{\text{CON}}(t) = Qp(t, T) \ .$$

- (iii) Put-call parity for digital AON options. *A buy and hold portfolio composed by a long digital AON call option and a long digital AON put option replicates a long position in the stock; i.e.*

$$C_{\text{AON}}(T) + P_{\text{AON}}(T) = S(T) \ .$$

Therefore, the prices $C_{\text{AON}}(t)$ and $P_{\text{AON}}(t)$ at time $t < T$ of the two options satisfy

$$C_{\text{AON}}(t) + P_{\text{AON}}(t) = S(t) \ .$$

- (iv) Put-call parity for plain vanilla options. *A buy and hold portfolio composed by a long plain vanilla call option and a short plain vanilla put option replicates a long forward. That is:*

$$C(T) - P(T) = F(T) \ .$$

Therefore, at time $t < T$, the prices $C(t)$ and $P(t)$ of the two options and of the long forward contract satisfy

$$C(t) - P(t) = F(t) \ .$$

- (v) Plain vanilla call as portfolio of digital calls. *A buy and hold portfolio composed by a long AON call and a short CON call, with cash amount $Q = K$, replicates a plain vanilla call option. That is:*

$$C(T) = C_{\text{AON}}(T) - C_{\text{CON}}(T) \ .$$

Therefore, the prices $C(t)$, $C_{\text{AON}}(t)$, and $C_{\text{CON}}(t)$ at time $t < T$ of the three options are related by

$$C(t) = C_{\text{AON}}(t) - C_{\text{CON}}(t) \ .$$

- (vi) Plain vanilla put as portfolio of digital puts: *A buy and hold portfolio composed by a long CON put and a short AON put, with cash amount $Q = K$, replicates a plain vanilla put option. That is:*

$$P(T) = P_{\text{CON}}(T) - P_{\text{AON}}(T) \ .$$

Therefore, for every time $t < T$, the prices $P(t)$, $P_{\text{CON}}(t)$, and $P_{\text{AON}}(t)$ satisfy

$$P(t) = P_{\text{CON}}(t) - P_{\text{AON}}(t) \ .$$

Proof.

- (i) By definition $F(T) = S(T) - K$, and this holds the replication statement. By no arbitrage, the replicating portfolio must have the same price in $t < T$ as the long forward contract, and hence the relation between prices in t holds true.
- (ii) By adding the two graphs in figure 6.1(d) and figure 6.1(g) we obtain a straight horizontal half line at level Q , i.e. the payoff of Q unit ZCBs. This can also be verified analytically:

$$C_{\text{CON}}(T) + P_{\text{CON}}(T) = Q\mathbb{1}_{\{S(T) > K\}} + Q\mathbb{1}_{\{S(T) \leq K\}} = Q[\mathbb{1}_{\{S(T) > K\}} + \mathbb{1}_{\{S(T) \leq K\}}] = Q .$$

The relation between prices in $t < T$ comes from no arbitrage.

- (iii) The proof of this item is left as an exercise (see exercise 1).
- (iv) By subtracting from the graph in figure 6.1(f) the graph in figure 6.1(i) we obtain a straight line with slope 45 degree and starting from $(0, -K)$, i.e. the payoff of a long forward (compare with the graph in figure 6.1(c)). This can also be verified analytically:

$$\begin{aligned} C(T) - P(T) &= [S(T) - K]^+ - [K - S(T)]^+ = \begin{cases} S(T) - K - 0 & \text{if } S(T) > K, \\ 0 - [K - S(T)] & \text{if } S(T) \leq K \end{cases} \\ &= S(T) - K = F(T) . \end{aligned}$$

The relation between prices in $t < T$ comes from no arbitrage.

- (v) By setting $Q = K$ and subtracting from the graph in figure 6.1(e) the graph in figure 6.1(d), we obtain the graph in figure 6.1(f), i.e. the payoff of a plain vanilla call. This can also be verified analytically:

$$\begin{aligned} C_{\text{AON}}(T) - C_{\text{CON}}(T) &= S(T)\mathbb{1}_{\{S(T) > K\}} - K\mathbb{1}_{\{S(T) > K\}} = [S(T) - K]\mathbb{1}_{\{S(T) > K\}} \\ &= [S(T) - K]^+ = C(T) . \end{aligned}$$

The relation between prices in $t < T$ comes from no arbitrage.

- (vi) The proof of this item is left as an exercise (see exercise 2 and exercise 3). □

Parity relation are fundamental both in pricing (as we are going to see in the next sections) and in hedging.

6.1.2 Moneyness

Definition 6.2 (Moneyness). A call option or a long forward contract with maturity T and strike K is said to be *in the money* (ITM) at time $t < T$ if $S(t) > K$, i.e. if it is going to pay a positive terminal payoff if the underlying does not change by the time the contract expires. If $S(t) < K$ the contract is said to be *out of the money* (OTM), because it has zero (in the case of a call) or negative (in the case of the forward) payoff under the same assumption. In the limit case of $S(t) = K$, the contract is said to be *at the money* (ATM). The *moneyness* of the contract is defined as the ratio

$$M(t) = \frac{S(t)}{K} ,$$

and a call option or a long forward contract is in, at, or out of the money at time t , depending on whether $M(t) > 1$, $M(t) = 1$ or $M(t) < 1$.

Symmetrically, a put option or a short forward contract is in, out, or at the money at time t if $S(t) < K$, $S(t) = K$ or $S(t) > K$, i.e. if $M(t) < 1$, $M(t) = 1$, or $M(t) > 1$.

Definition 6.3 (Forward moneyness). A call option or a long forward contract with maturity T and strike K is said to be *in the money forward* (ITMF) at time $t < T$ if $S(t) > Kp(t, T)$, i.e. if it is going to pay a positive terminal payoff if the underlying asset grows exactly at the risk free rate until maturity. If $S(t) < Kp(t, T)$ or $S(t) = Kp(t, T)$ the contract is said to be *out of the money forward* (OTMF) or *at the money forward* (ATMF). The *forward moneyness* of the contract is defined as the ratio

$$M_F(t) = \frac{S(t)}{Kp(t, T)} ,$$

and a call option or a long forward contract is in, at, or out of the money forward at time t , depending on whether $M_F(t) > 1$, $M_F(t) = 1$, or $M_F(t) < 1$.

Symmetrically, a put option or a short forward contract is in, out, or at the money forward at time t if $S(t) < Kp(t, T)$, $S(t) = Kp(t, T)$, or $S(t) > Kp(t, T)$, i.e. if $M_F(t) < 1$, $M_F(t) = 1$, or $M_F(t) > 1$.

Theorem 6.4 (At the money forward pricing). *In the at the money forward case,*

$$(i) \quad F(t) = 0.$$

$$(ii) \quad C(t) = P(t).$$

Proof. Follows trivially from theorem 6.1(ii) and (iv). $\square$

Remark 6.5. The definition of moneyness is not completely standard, either in literature or in operational practice. Some distinguish between moneyness in the call sense, defined as $M(t)$, and call moneyness in the put sense the reciprocal $1/M(t)$. Others define moneyness as what we have called forward moneyness. $\square$

6.2 European option pricing in the Black & Scholes model

In this section we assume the standard Black & Scholes model, i.e. the underlying follows a Geometric Brownian Motion under $\mathbb{Q}$, with constant risk-free rate r and constant volatility $\sigma > 0$. Let us recall that the probability distribution function (PDF), also called density, of a standard normal random variable Z is

$$f : \mathbb{R} \rightarrow \mathbb{R}^+ , \quad f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} , \quad \forall z \in \mathbb{R} .$$

The corresponding cumulative distribution function (CDF) is denoted by N and is

$$N : \mathbb{R} \rightarrow [0, 1] , \quad N(z) = \mathbb{Q}(Z < z) = \int_{-\infty}^z f(u) du , \quad \forall z \in \mathbb{R} .$$

Notice that $f(z) = N'(z)$, for all $z \in \mathbb{R}$.

Lemma 6.6. *For every $z \in \mathbb{R}$, $N(z) = 1 - N(-z)$.*

Proof. Recalling that N' is a PDF and hence integrates to 1 over the whole real line, and using the domain decomposition property of integrals, we have

$$N(z) = \int_{-\infty}^z N'(u) du = \int_{-\infty}^{+\infty} N'(u) du - \int_z^{+\infty} N'(u) du = 1 - \int_z^{+\infty} N'(u) du .$$

Let us now apply a change of variable in the last integral and set $v = -u$. Therefore $du = -dv$, the upper integration limit $u = +\infty$ becomes $v = -\infty$ and the lower integration limit $u = z$ becomes $v = -z$. Therefore

$$\int_z^{+\infty} N'(u) du = - \int_{-z}^{-\infty} N'(-v) dv = \int_{-\infty}^{-z} N'(-v) dv = \int_{-\infty}^{-z} N'(v) dv ,$$

where in the last equality we used the fact that N' is an even function, i.e. $N'(-v) = N'(v)$ for every $v \in \mathbb{R}$. Hence

$$N(z) = 1 - \int_{-\infty}^{-z} N'(v) dv = 1 - N(-z) . \quad \square$$

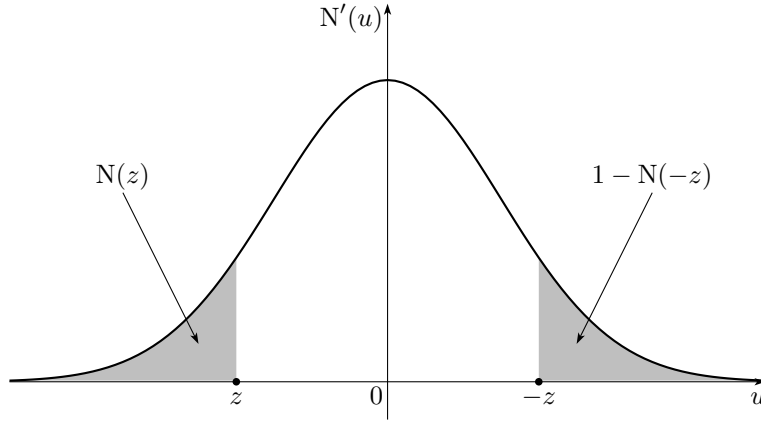


Figure 6.2. Graphical representation of $N(z) = 1 - N(-z)$.

Figure 6.2 depicts graphically the statement of lemma 6.6.

6.2.1 Digital CON option pricing

Theorem 6.7 (Digital CON call option pricing). *In the Black & Scholes model, the price at time $t < T$ of a digital CON call option with maturity T , strike K , and cash amount Q is given by¹*

$$C_{\text{CON}}(t) = Q N(d_2) e^{-r(T-t)} , \quad \text{where} \quad d_2 = \frac{\log \frac{S(t)}{K} + (r - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} .$$

Proof. We know that

$$\begin{aligned} C_{\text{CON}}(t) &= \mathbb{E}_t^{\mathbb{Q}} [C_{\text{CON}}(T) e^{-r(T-t)}] = \mathbb{E}_t^{\mathbb{Q}} [Q \mathbb{1}_{\{S(T) > K\}}] e^{-r(T-t)} \\ &= Q \mathbb{Q}(S(T) > K | \mathcal{F}_t^S) e^{-r(T-t)} = Q[1 - \mathbb{Q}(S(T) \leq K | \mathcal{F}_t^S)] e^{-r(T-t)} . \end{aligned}$$

We also know that, under the $\mathbb{Q}$ measure

$$S(T) = S(t) e^{(r - \frac{1}{2}\sigma^2)(T-t) + \sigma[W(T) - W(t)]} .$$

Hence, the inequality $S(T) \leq K$ is equivalent to

$$\begin{aligned} S(t) e^{(r - \frac{1}{2}\sigma^2)(T-t) + \sigma[W(T) - W(t)]} &\leq K , \\ \log \frac{S(t)}{K} + \left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma[W(T) - W(t)] &\leq 0 , \\ \frac{\log \frac{S(t)}{K} + (r - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} &\leq -\frac{1}{\sqrt{T-t}}[W(T) - W(t)] , \\ -d_2 &\geq \frac{1}{\sqrt{T-t}}[W(T) - W(t)] . \end{aligned}$$

Now, conditional to $\mathcal{F}_t^S$ and under the $\mathbb{Q}$ measure, $W(T) - W(t) \sim N(0, T-t)$, hence $Z = [W(T) - W(t)]/\sqrt{T-t} \sim N(0, 1)$. Therefore,

$$\mathbb{Q}(S(T) \leq K | \mathcal{F}_t^S) = \mathbb{Q}(Z \leq -d_2 | \mathcal{F}_t^S) = N(-d_2) ,$$

and

$$C_{\text{CON}}(t) = Q[1 - N(-d_2)] e^{-r(T-t)} = Q N(d_2) e^{-r(T-t)} . \quad \square$$

¹Notation d_2 is traditional in financial literature and dates back to the original work of Black and Scholes. As we are going to see in theorem 6.10, there is also a d_1 , strongly connected with d_2 , and both appear in the Black and Scholes formula (theorem 6.13).

Corollary 6.8 (Digital CON put option pricing). *In the Black & Scholes model, the price at time $t < T$ of a digital CON put option with maturity T , strike K , and cash amount Q is given by*

$$P_{\text{CON}}(t) = Q N(-d_2) e^{-r(T-t)} .$$

Proof. By combining theorem 6.1(ii), the previous theorem, and lemma 6.6;

$$\begin{aligned} P_{\text{CON}}(t) &= -C_{\text{CON}}(t) + Q e^{-r(T-t)} = -Q N(d_2) e^{-r(T-t)} + Q e^{-r(T-t)} = Q[1 - N(d_2)] e^{-r(T-t)} \\ &= Q N(-d_2) e^{-r(T-t)} . \end{aligned} \quad \square$$

Remark 6.9. The formula for d_2 can be also written using moneyness or forward moneyness:

$$d_2 = \frac{\log M(t) + (r - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} = \frac{\log M_F(t) - \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}} .$$

As a consequence, in the ATMF case, $d_2 = -\frac{1}{2}\sigma\sqrt{T-t}$. $\square$

6.2.2 Digital AON option pricing

Theorem 6.10 (Digital AON call option pricing). *In the Black & Scholes model, the price at time $t < T$ of a digital AON call option with maturity T and strike K is given by*

$$C_{\text{AON}}(t) = S(t) N(d_1) , \quad \text{where} \quad d_1 = \frac{\log \frac{S(t)}{K} + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} = d_2 + \sigma\sqrt{T-t} .$$

Proof. We know that

$$\begin{aligned} C_{\text{AON}}(t) &= \mathbb{E}_t^{\mathbb{Q}} [C_{\text{AON}}(T) e^{-r(T-t)}] = \mathbb{E}_t^{\mathbb{Q}} [S(T) \mathbb{1}_{\{S(T) > K\}} e^{-r(T-t)}] \\ &= \mathbb{E}_t^{\mathbb{Q}} [S(t) e^{(r - \frac{1}{2}\sigma^2)(T-t) + \sigma[W(T) - W(t)]} \mathbb{1}_{\{S(T) > K\}} e^{-r(T-t)}] \\ &= S(t) \mathbb{E}_t^{\mathbb{Q}} [e^{-\frac{1}{2}\sigma^2(T-t) + \sigma[W(T) - W(t)]} \mathbb{1}_{\{S(T) > K\}}] . \end{aligned}$$

Now, as in the proof of theorem 6.7, under the $\mathbb{Q}$ measure and conditional to $\mathcal{F}_t^S$, $W(T) - W(t) = \sqrt{T-t} Z$, where $Z \sim N(0, 1)$ and the inequality $S(T) > K$ is equivalent to $Z > -d_2$. Therefore

$$\begin{aligned} \mathbb{E}_t^{\mathbb{Q}} [e^{-\frac{1}{2}\sigma^2(T-t) + \sigma[W(T) - W(t)]} \mathbb{1}_{\{S(T) > K\}}] &= \int_{-\infty}^{+\infty} e^{-\frac{1}{2}\sigma^2(T-t) + \sigma\sqrt{T-t}z} \mathbb{1}_{\{z > -d_2\}} N'(z) dz \\ &= \int_{-d_2}^{+\infty} e^{-\frac{1}{2}\sigma^2(T-t) + \sigma\sqrt{T-t}z} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\ &= \int_{-d_2}^{+\infty} e^{-\frac{1}{2}\sigma^2(T-t) + \sigma\sqrt{T-t}z - z^2/2} \frac{1}{\sqrt{2\pi}} dz \end{aligned}$$

Now,

$$-\frac{1}{2}\sigma^2(T-t) + \sigma\sqrt{T-t}z - z^2/2 = -\frac{1}{2}(z - \sigma\sqrt{T-t})^2 .$$

Consider now the change of variable $x = z - \sigma\sqrt{T-t}$; then $dx = dz$, $z = +\infty$ translates to $x = +\infty$ and $z = -d_2$ becomes $x = -d_2 - \sigma\sqrt{T-t} = -d_1$. Hence

$$\begin{aligned} \int_{-d_2}^{+\infty} e^{-\frac{1}{2}\sigma^2(T-t) + \sigma\sqrt{T-t}z - z^2/2} \frac{1}{\sqrt{2\pi}} dz &= \int_{-d_1}^{+\infty} e^{-\frac{1}{2}x^2} \frac{1}{\sqrt{2\pi}} dx \\ &= 1 - \int_{-\infty}^{-d_1} e^{-\frac{1}{2}x^2} \frac{1}{\sqrt{2\pi}} dx = 1 - N(-d_1) = N(d_1) , \end{aligned}$$

where in the last step we used lemma 6.6. Putting all pieces together, we get

$$C_{\text{AON}}(t) = S(t) N(d_1) . \quad \square$$

Corollary 6.11 (Digital AON put option pricing). *In the Black & Scholes model, the price at time $t < T$ of a digital AON put option with maturity T and strike K is given by*

$$P_{\text{AON}}(t) = S(t) N(-d_1) .$$

Proof. By combining theorem 6.1(iii), the previous theorem, and lemma 6.6:

$$P_{\text{AON}}(t) = -C_{\text{AON}}(t) + S(t) = -S(t) N(d_1) + S(t) = S(t)[1 - N(d_1)] = S(t) N(-d_1) . \quad \square$$

Remark 6.12. The formula for d_1 can be also written using moneyness or forward moneyness:

$$d_1 = \frac{\log M(t) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} = \frac{\log M_F(t) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}} .$$

As a consequence, in the ATMF case, $d_1 = \frac{1}{2}\sigma\sqrt{T-t}$. In particular, in this case $d_1 = -d_2$. $\square$

6.2.3 Plain vanilla option pricing

Theorem 6.13 (The Black & Scholes formula for the plain vanilla call option). *In the Black & Scholes model, the price at time $t < T$ of a plain vanilla call option with maturity T and strike K is given by*

$$C(t) = S(t) N(d_1) - K e^{-r(T-t)} N(d_2) .$$

Proof. Combining the results of theorem 6.1(v), of theorem 6.10, and of theorem 6.7, we have:

$$C(t) = C_{\text{AON}}(t) - C_{\text{CON}}(t) = S(t) N(d_1) - K e^{-r(T-t)} N(d_2) . \quad \square$$

Corollary 6.14 (The Black & Scholes formula for the plain vanilla put option). *In the Black & Scholes model, the price at time $t < T$ of a plain vanilla put option with maturity T and strike K is given by*

$$P(t) = K e^{-r(T-t)} N(-d_2) - S(t) N(-d_1) .$$

Proof. Combining the results of theorem 6.1(iv), of theorem 6.1(i), and of theorem 6.13, we get:

$$\begin{aligned} P(t) &= C(t) - F(t) = S(t) N(d_1) - K e^{-r(T-t)} N(d_2) - S(t) + K e^{-r(T-t)} \\ &= -S(t)[1 - N(d_1)] + K e^{-r(T-t)}[1 - N(d_2)] = K e^{-r(T-t)} N(-d_2) - S(t) N(-d_1) , \end{aligned}$$

where in the last step we used lemma 6.6. $\square$

Remark 6.15. In the ATMF case, where $d_2 = -d_1$, we can simplify the Black & Scholes formulae for plain vanilla options:

$$C(t) = S(t)[2N(d_1) - 1] = P(t). \quad \square$$

6.3 The Greeks

Consider a T -claim (non necessarily a simple one) and let $f(t, s)$ be its pricing function. The (partial) derivatives of the pricing function are a relevant information, as they measure how the price moves when the independent variables t and s change. Furthermore, the pricing function depends also on the model parameters (e.g., in the Black and Scholes model, on volatility σ) and on the contractual parameters (e.g., for an option, on the strike K); therefore partial derivatives with respect to these parameters are also to be considered. These derivatives are traditionally denoted using Greek letters, and are therefore called *the Greeks* of the contingent claim. The main Greeks are:

Delta: The derivative with respect to the price of the underlying. It is denoted by Δ and measures the change of the derivative price due to a change of 1 unit of account of the underlying price.

Gamma: The second derivative with respect to the price of the underlying, i.e. the derivative of Δ with respect to s . It is denoted by Γ and represents the change in Δ induced by a change of 1 unit in s .

Vega: The derivative with respect to volatility. It is denoted by $\mathcal{V}$ and measures the change of the derivative price with respect to a change of 1 unit ($= +100\%$) in volatility.² Since a unit change in volatility is *very* extreme, $\mathcal{V}$ is often divided by 100 in order to represent the change relative to $+1\%$ in volatility. This Greek makes sense, at least with this definition, only in models where volatility is a constant, such as the Black and Scholes model.

Theta: The derivative with respect to time t , denoted by Θ . It measures the so called *time decay*, the sensitivity of the derivative price with respect to time.

Rho: The derivative with respect to the short rate. Denoted by ρ , it makes sense, at least with this definition, only for models where the short rate is assumed to be constant. It represents the change in the derivative price to a unit change ($= +100\%$) in the short rate. Often it is divided by 100, to measure the sensitivity to a change of $+1\%$ in the short rate.

Other Greeks can be defined (e.g. Vanna, Volga, Charm, ...), and some of them are particularly relevant for specific claims.

6.3.1 Greeks in the Black and Scholes model

Assuming a standard Black and Scholes model, the five main Greeks are

$$\Delta = \frac{\partial f}{\partial s} , \quad \Gamma = \frac{\partial^2 f}{\partial s^2} = \frac{\partial \Delta}{\partial s} , \quad \mathcal{V} = \frac{\partial f}{\partial \sigma} , \quad \Theta = \frac{\partial f}{\partial t} , \quad \rho = \frac{\partial f}{\partial r}$$

(to simplify exposition we omit functional dependencies). We are going to compute them for the basic European contingent claims. Unlike what was done in the previous section, it will be easier to first calculate the Greeks for the plain vanilla options, and then obtain those of the digital options by exploiting the parity relationships.

Let's start with the Greeks of the simplest T -claims.

Theorem 6.16 (Greeks of the ZCB). *The Greeks at time t of a unit ZCB with time to maturity $\tau = T - t$ are*

$$\Delta_{\text{ZCB}} = \Gamma_{\text{ZCB}} = \mathcal{V}_{\text{ZCB}} = 0 , \quad \Theta_{\text{ZCB}} = r e^{-r\tau} , \quad \rho_{\text{ZCB}} = -\tau e^{-r\tau} .$$

Proof. The computation of the partial derivatives of the ZCB pricing function $f = e^{-r\tau}$ is straightforward and is left as an exercise (exercise 13). $\square$

Theorem 6.17 (Greeks of the long forward contract). *The Greeks at time t of a (long) forward contract with time to maturity $\tau = T - t$ and strike K are*

$$\Delta_F = 1 , \quad \Gamma_F = \mathcal{V}_F = 0 , \quad \Theta_F = -rK e^{-r\tau} , \quad \rho_F = \tau K e^{-r\tau} .$$

Proof. The computation of the partial derivatives of the forward pricing function $f = s - K e^{-r\tau}$ is straightforward and is left as an exercise (exercise 14). $\square$

Greeks of plain vanilla options

In this subsection we consider plain vanilla options at time t , with maturity in T , strike K , volatility σ and time to maturity $\tau = T - t$. It is worthwhile first of all to demonstrate a remarkable lemma.

²Nonetheless, in the Greek alphabet there is no letter called “Vega” or written with $\mathcal{V}$.

Lemma 6.18 (Zero lemma). *For any $s, K > 0$, $\tau, r, \sigma \in \mathbb{R}$*

$$s N'(d_1) - K e^{-r\tau} N'(d_2) = 0 ,$$

where d_1 and d_2 are the standard Black and Scholes “d”s.

Proof. The statement holds if and only if

$$s N'(d_1) = K e^{-r\tau} N'(d_2) \iff \frac{s}{K} e^{r\tau} = \frac{N'(d_2)}{N'(d_1)} \iff \log \frac{s}{K} + r\tau = \frac{d_1^2 - d_2^2}{2} .$$

Now, the right hand side of the last condition is

$$\begin{aligned} \frac{d_1^2 - d_2^2}{2} &= \frac{1}{2}(d_1 + d_2)(d_1 - d_2) = \frac{1}{2}(2d_1 - \sigma\sqrt{\tau})\sigma\sqrt{\tau} = \log \frac{s}{K} + (r + \frac{1}{2}\sigma^2)\tau - \frac{1}{2}\sigma^2\tau \\ &= \log \frac{s}{K} + r\tau \end{aligned}$$

and this completes the proof. $\square$

The Zero lemma states that if we write the Black and Scholes call formula using the PDF N' instead of the CDF N , we get always zero. This may happen also by mistake, as in many software packages the name for the two functions is often similar and sometimes confusing.³

Some useful computations:

Lemma 6.19 (Identities between the derivatives of the B&S “d”s). *For any $s, K, \tau, r, \sigma \in \mathbb{R}$, with $s > 0$, $K > 0$, $\tau > 0$, and $\sigma > 0$,*

$$\frac{\partial d_1}{\partial s} = \frac{\partial d_2}{\partial s} , \tag{6.1}$$

$$\frac{\partial d_1}{\partial \sigma} = \frac{\partial d_2}{\partial \sigma} + \sqrt{\tau} , \tag{6.2}$$

$$\frac{\partial d_1}{\partial \tau} = \frac{\partial d_2}{\partial \tau} + \frac{\sigma}{2\sqrt{\tau}} , \tag{6.3}$$

$$\frac{\partial d_1}{\partial r} = \frac{\partial d_2}{\partial r} . \tag{6.4}$$

Proof. Straightforward computations, left as an exercise (exercise 15). $\square$

Lemma 6.20 (Derivatives of the B&S “d”s). *For any $s, K, \tau, r, \sigma \in \mathbb{R}$, with $s > 0$, $K > 0$, $\tau > 0$, and $\sigma > 0$,*

$$\frac{\partial d_1}{\partial s} = \frac{1}{s\sigma\sqrt{\tau}} , \tag{6.5}$$

$$\frac{\partial d_1}{\partial \sigma} = -\frac{d_2}{\sigma} , \quad \frac{\partial d_2}{\partial \sigma} = -\frac{d_1}{\sigma} , \tag{6.6}$$

$$\frac{\partial d_1}{\partial \tau} = -\frac{d_2}{2\tau} + \frac{r}{\sigma\sqrt{\tau}} , \quad \frac{\partial d_2}{\partial \tau} = -\frac{d_1}{2\tau} + \frac{r}{\sigma\sqrt{\tau}} , \tag{6.7}$$

$$\frac{\partial d_1}{\partial r} = \frac{\sqrt{\tau}}{\sigma} . \tag{6.8}$$

Proof. Straightforward computations, left as an exercise (exercise 16). $\square$

Theorem 6.21 (Greeks of the Plain Vanilla Call). *For a plain vanilla call option we have:*

$$\begin{aligned} \Delta_C &= N(d_1) , & \Gamma_C &= \frac{N'(d_1)}{s\sigma\sqrt{\tau}} , & \mathcal{V}_C &= s\sqrt{\tau} N'(d_1) , \\ \Theta_C &= -\frac{s\sigma N'(d_1)}{2\sqrt{\tau}} - rK e^{-r\tau} N(d_2) , & \rho_C &= \tau K e^{-r\tau} N(d_2) . \end{aligned}$$

³For example, in the Excel spreadsheet, both N and N' are computed by the same Excel function, called with the last optional argument set to **true** (default) to compute the former, to **false** to compute the latter.

Proof. All the Greeks are computed by deriving the B&S formula, then applying identities of lemma 6.19, the Zero lemma, and, for Γ_C , also lemma 6.20:

$$\begin{aligned}\Delta_C &= \frac{\partial C}{\partial s} = \frac{\partial}{\partial s} [s N(d_1) - K e^{-r\tau} N(d_2)] \\ &= N(d_1) + s N'(d_1) \frac{\partial d_1}{\partial s} - K e^{-r\tau} N'(d_2) \frac{\partial d_2}{\partial s} \\ &= N(d_1) + s N'(d_1) \frac{\partial d_2}{\partial s} - K e^{-r\tau} N'(d_2) \frac{\partial d_2}{\partial s} \quad (\text{by (6.1)}) \\ &= N(d_1) + \frac{\partial d_2}{\partial s} [s N'(d_1) - K e^{-r\tau} N'(d_2)] = N(d_1) \quad (\text{by the Zero lemma});\end{aligned}$$

$$\Gamma_C = \frac{\partial^2 C}{\partial s^2} = \frac{\partial \Delta_C}{\partial s} = \frac{\partial N(d_1)}{\partial s} = N'(d_1) \frac{\partial d_1}{\partial s} = \frac{N'(d_1)}{s\sigma\sqrt{\tau}} \quad (\text{by (6.5)});$$

$$\begin{aligned}\mathcal{V}_C &= \frac{\partial C}{\partial \sigma} = \frac{\partial}{\partial \sigma} [s N(d_1) - K e^{-r\tau} N(d_2)] \\ &= s N'(d_1) \frac{\partial d_1}{\partial \sigma} - K e^{-r\tau} N'(d_2) \frac{\partial d_2}{\partial \sigma} \\ &= s N'(d_1) \left(\frac{\partial d_2}{\partial \sigma} + \sqrt{\tau} \right) - K e^{-r\tau} N'(d_2) \frac{\partial d_2}{\partial \sigma} \quad (\text{by (6.2)}) \\ &= s\sqrt{\tau} N'(d_1) + \frac{\partial d_2}{\partial \sigma} [s N'(d_1) - K e^{-r\tau} N'(d_2)] = s\sqrt{\tau} N'(d_1) \quad (\text{by the Zero lemma}).\end{aligned}$$

To compute Θ_C , notice that $\partial\tau/\partial t = -1$ and therefore

$$\Theta_C = \frac{\partial C}{\partial t} = \frac{\partial C}{\partial \tau} \frac{\partial \tau}{\partial t} = -\frac{\partial C}{\partial \tau}$$

by the chain rule. Hence:

$$\begin{aligned}\Theta_C &= -\frac{\partial C}{\partial \tau} = -\frac{\partial}{\partial \tau} [s N(d_1) - K e^{-r\tau} N(d_2)] \\ &= -s N'(d_1) \frac{\partial d_1}{\partial \tau} - r K e^{-r\tau} N(d_2) + K e^{-r\tau} N'(d_2) \frac{\partial d_2}{\partial \tau} \\ &= -s N'(d_1) \left(\frac{\partial d_2}{\partial \tau} + \frac{\sigma}{2\sqrt{\tau}} \right) + K e^{-r\tau} N'(d_2) \frac{\partial d_2}{\partial \tau} - r K e^{-r\tau} N(d_2) \quad (\text{by (6.3)}) \\ &= -\frac{\partial d_2}{\partial \tau} [s N'(d_1) - K e^{-r\tau} N'(d_2)] - \frac{s\sigma N'(d_1)}{2\sqrt{\tau}} - r K e^{-r\tau} N(d_2) \\ &= -\frac{s\sigma N'(d_1)}{2\sqrt{\tau}} - r K e^{-r\tau} N(d_2) \quad (\text{by the Zero lemma}).\end{aligned}$$

The proof of the formula for ρ_C is left as an exercise (exercise 24). $\square$

Figure 6.3 graphs the price and the Greeks of a plain vanilla call as functions of s .

Corollary 6.22 (Greeks of a Plain Vanilla Put). *For a plain vanilla put option we have:*

$$\begin{aligned}\Delta_P &= \Delta_C - 1 = -N(-d_1) , & \Gamma_P &= \Gamma_C , & \mathcal{V}_P &= \mathcal{V}_C , \\ \Theta_P &= \Theta_C + r K e^{-r\tau} = -\frac{s\sigma N'(d_1)}{2\sqrt{\tau}} + r K e^{-r\tau} N(-d_2) , \\ \rho_P &= \rho_C - \tau K e^{-r\tau} = -\tau K e^{-r\tau} N(-d_2) .\end{aligned}$$

Proof. All the Greeks are computed using put-call parity to link them to the Greek of the call of theorem 6.21 and to the Greeks of the long forward contract (theorem 6.17). The second equalities in the formulae for Δ_P , Θ_P , and ρ_P are obtained by simple manipulations, using lemma 6.6. The details are left as an exercise (exercise 22). $\square$

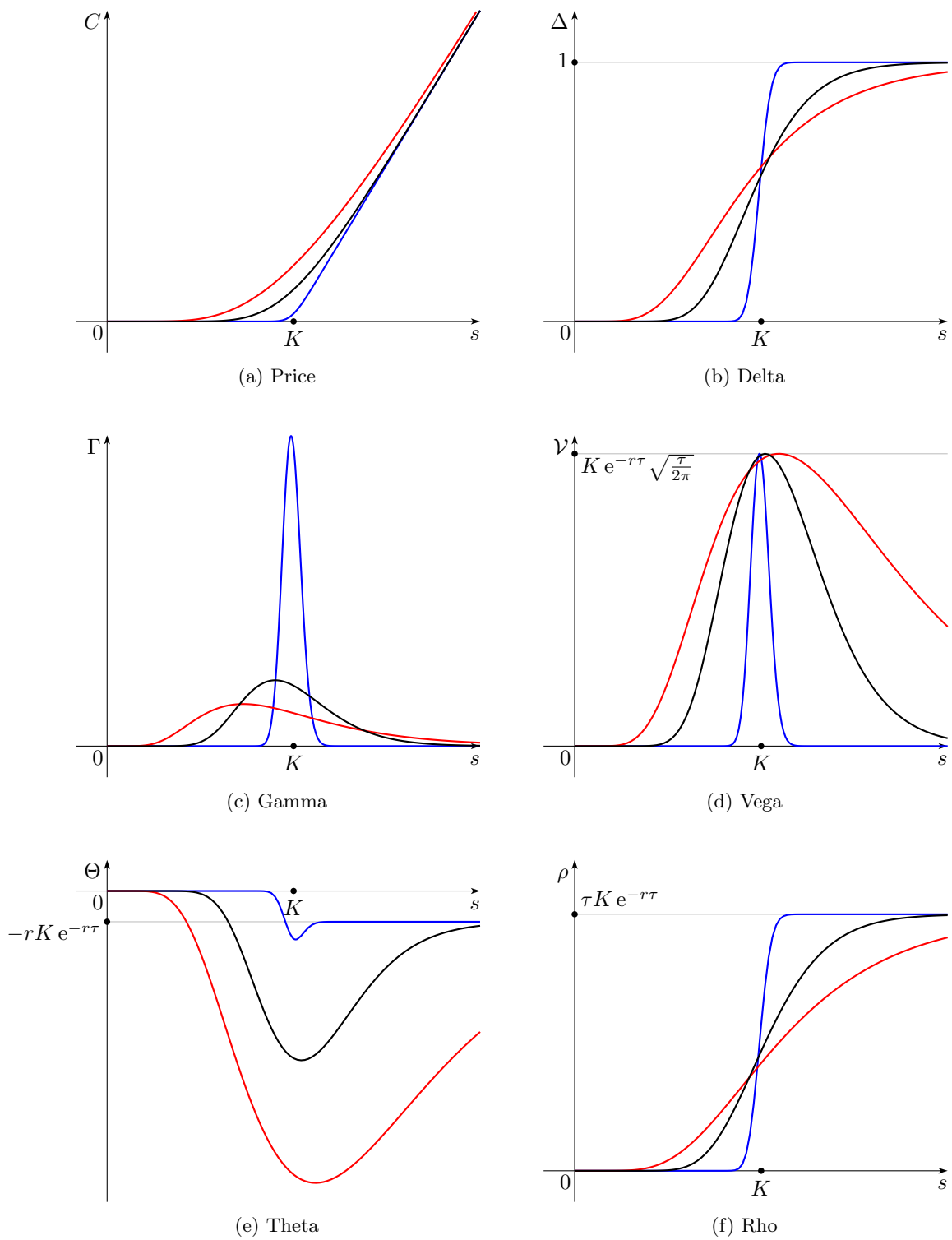


Figure 6.3. Price and Greeks of a p.v. call as functions of s : $K = 100$, $\sigma = 5\%$ (blue), 25% (black), 45% (red), $r = 1\%$, $\tau = 1$.

Remark 6.23. For plain vanilla options:

- Delta is positive in the case of a call, negative in the case of a put.
- Call and put have the same positive Gamma.
- Call and put have the same positive Vega.

□

Remark 6.24. Using the Zero lemma, in the formulae for the Greeks we can replace:

$$N'(d_1) \quad \text{with} \quad \frac{K}{s} e^{-r\tau} N'(d_2) , \quad \text{and} \quad N'(d_2) \quad \text{with} \quad \frac{s}{K e^{-r\tau}} N'(d_1) ,$$

to obtain different (but equivalent) formulae. Some examples are:

$$\begin{aligned} \Gamma_C &= \Gamma_P = \frac{K e^{-r\tau} N'(d_2)}{s^2 \sigma \sqrt{\tau}} , \\ \mathcal{V}_C &= \mathcal{V}_P = \sqrt{\tau} K e^{-r\tau} N'(d_2) , \\ \Theta_C &= -K e^{-r\tau} \left[\frac{\sigma N'(d_2)}{2\sqrt{\tau}} + r N(d_2) \right] , \\ \Theta_P &= -K e^{-r\tau} \left[\frac{\sigma N'(d_2)}{2\sqrt{\tau}} - r N(-d_2) \right] . \end{aligned}$$

□

Greeks of digital CON options

In this subsection we consider CON options at time t , with cash amount Q , maturity in T , strike K , volatility σ and time to maturity $\tau = T - t$.

Theorem 6.25 (Greeks of the Digital CON Call). *For a digital CON call option we have:*

$$\begin{aligned} \Delta_{C_{\text{CON}}} &= \frac{Q e^{-r\tau} N'(d_2)}{s \sigma \sqrt{\tau}} , \quad \Gamma_{C_{\text{CON}}} = -\frac{d_1 Q e^{-r\tau} N'(d_2)}{s^2 \sigma^2 \tau} , \quad \mathcal{V}_{C_{\text{CON}}} = -\frac{d_1 Q e^{-r\tau} N'(d_2)}{\sigma} , \\ \Theta_{C_{\text{CON}}} &= Q e^{-r\tau} \left[\frac{d_1 \sigma \sqrt{\tau} - 2r\tau}{2\sigma \tau \sqrt{\tau}} N'(d_2) + r N(d_2) \right] , \\ \rho_{C_{\text{CON}}} &= Q e^{-r\tau} \left[\frac{\sqrt{\tau} N'(d_2)}{\sigma} - \tau N(d_2) \right] . \end{aligned}$$

Proof. The proof requires some tedious calculations, but is straightforward and therefore left as an exercise (exercise 27). □

Corollary 6.26 (Greeks of the Digital CON Put). *For a digital CON put option we have:*

$$\begin{aligned} \Delta_{P_{\text{CON}}} &= -\Delta_{C_{\text{CON}}} , \quad \Gamma_{P_{\text{CON}}} = -\Gamma_{C_{\text{CON}}} , \quad \mathcal{V}_{P_{\text{CON}}} = -\mathcal{V}_{C_{\text{CON}}} , \\ \Theta_{P_{\text{CON}}} &= -\Theta_{C_{\text{CON}}} + rQ e^{-r\tau} = -Q e^{-r\tau} \left[\frac{d_1 \sigma \sqrt{\tau} - 2r\tau}{2\sigma \tau \sqrt{\tau}} N'(d_2) - r N(-d_2) \right] , \\ \rho_{P_{\text{CON}}} &= -\rho_{C_{\text{CON}}} - \tau Q e^{-r\tau} = -Q e^{-r\tau} \left[\frac{\sqrt{\tau} N'(d_2)}{\sigma} + \tau N(-d_2) \right] . \end{aligned}$$

Proof. Using the CON put call parity (theorem 6.1(ii)) and, where needed, lemma 6.6 to simplify formulae, we get

$$\begin{aligned} \Delta_{P_{\text{CON}}} &= \frac{\partial P_{\text{CON}}}{\partial s} = -\frac{\partial C_{\text{CON}}}{\partial s} + \frac{\partial}{\partial s} (Q e^{-r\tau}) = -\Delta_{C_{\text{CON}}} ; \\ \Gamma_{P_{\text{CON}}} &= \frac{\partial^2 P_{\text{CON}}}{\partial s^2} = -\frac{\partial^2 C_{\text{CON}}}{\partial s^2} + \frac{\partial^2}{\partial s^2} (Q e^{-r\tau}) = -\Gamma_{C_{\text{CON}}} ; \\ \mathcal{V}_{P_{\text{CON}}} &= \frac{\partial P_{\text{CON}}}{\partial \sigma} = -\frac{\partial C_{\text{CON}}}{\partial \sigma} + \frac{\partial}{\partial \sigma} (Q e^{-r\tau}) = -\mathcal{V}_{C_{\text{CON}}} ; \\ \Theta_{P_{\text{CON}}} &= \frac{\partial P_{\text{CON}}}{\partial t} = -\frac{\partial C_{\text{CON}}}{\partial t} + \frac{\partial}{\partial t} (Q e^{-r\tau}) = -\Theta_{C_{\text{CON}}} + rQ e^{-r\tau} \end{aligned}$$

$$\begin{aligned}
&= -Q e^{-r\tau} \left[\frac{d_1 \sigma \sqrt{\tau} - 2r\tau}{2\sigma\tau\sqrt{\tau}} N'(d_2) - r N(-d_2) \right] ; \\
\rho_{P_{\text{CON}}} &= \frac{\partial P_{\text{CON}}}{\partial r} = -\frac{\partial C_{\text{CON}}}{\partial r} + \frac{\partial}{\partial r} (Q e^{-r\tau}) = -\rho_{C_{\text{CON}}} - \tau Q e^{-r\tau} \\
&= -Q e^{-r\tau} \left[\frac{\sqrt{\tau} N'(d_2)}{\sigma} + \tau N(-d_2) \right] . \quad \square
\end{aligned}$$

Greeks of digital AON options

The Greeks for digital AON call options can be easily computed by combining the parity relation 6.1(v), theorem 6.21, and theorem 6.25.

Theorem 6.27 (Greeks of the Digital AON Call). *For a digital AON call option we have:*

$$\begin{aligned}
\Delta_{C_{\text{AON}}} &= N(d_1) + \frac{N'(d_1)}{\sigma\sqrt{\tau}} , & \Gamma_{C_{\text{AON}}} &= -\frac{d_2 N'(d_1)}{s\sigma^2\tau} , & \mathcal{V}_{C_{\text{AON}}} &= -\frac{s d_2 N'(d_1)}{\sigma} , \\
\Theta_{C_{\text{AON}}} &= \frac{s(d_2 - 2r\tau) N'(d_1)}{2\sigma\tau\sqrt{\tau}} , & \rho_{C_{\text{AON}}} &= \frac{s\sqrt{\tau} N'(d_1)}{\sigma} .
\end{aligned}$$

Proof. The proof is left as an exercise (exercise 28). □

Corollary 6.28 (Greeks of the Digital AON Put). *For a digital AON put option the Greeks are:*

$$\begin{aligned}
\Delta_{P_{\text{AON}}} &= 1 - \Delta_{C_{\text{AON}}} = N(-d_1) - \frac{N'(d_1)}{\sigma\sqrt{\tau}} , & \Gamma_{P_{\text{AON}}} &= -\Gamma_{C_{\text{AON}}} , \\
\mathcal{V}_{P_{\text{AON}}} &= -\mathcal{V}_{C_{\text{AON}}} , & \Theta_{P_{\text{AON}}} &= -\Theta_{C_{\text{AON}}} , & \rho_{P_{\text{AON}}} &= -\rho_{C_{\text{AON}}} .
\end{aligned}$$

Proof. Follows directly from the AON put-call parity (theorem 6.1(iii)), theorem 6.27, and from the observation that the only non-zero Greek of the underlying is $\Delta_s = 1$. □

6.4 vba Implementation

This section is devoted to the vba implementation of the formulae for prices and Greeks obtained in the Black and Scholes model.

6.4.1 Helper subroutines

We start with some helper subroutines, to compute Black and Scholes “d”s. Functions d2() and d1() take as input both market data (current level of the underlying $s = s$, volatility $\sigma = \sigma$, and short rate $r = r$) and contractual data (strike $K = K$ and time to maturity $\tau = \tau$) and output $d2 = d_2$ and $d1 = d_1$.

Option Explicit

```

Function d2(s As Double, sigma As Double, r As Double, K As Double, _
    tau As Double) As Double
    Dim num As Double, den As Double
    num = Log(s / K) + (r - 0.5 * sigma ^ 2) * tau
    den = sigma * Sqr(tau)
    d2 = num / den
End Function

```

```

Function d1(s As Double, sigma As Double, r As Double, K As Double, _
    tau As Double) As Double
    d1 = d2(s, sigma, r, K, tau) + sigma * Sqr(tau)
End Function

```

Subroutine d1d2() computes both d_1 and d_2 in one step and is more efficient in the case we need both (as in the case of formulae of section 6.2.3). The first five arguments are the same inputs of functions d1() and d2(). The last two arguments of subroutine d1d2() are outputs: $dd1 = d_1$ and $dd2 = d_2$.

```
Sub d1d2(s As Double, sigma As Double, r As Double, K As Double, _
        tau As Double, dd1 As Double, dd2 As Double)
    dd2 = d2(s, sigma, r, K, tau)
    dd1 = dd2 + sigma * Sqr(tau)
End Sub
```

6.4.2 Option pricing

Digital CON options

The CON pricing functions are $\text{callCON} = C_{\text{CON}}$ for calls and $\text{putCON} = P_{\text{CON}}$ for puts. The two pricing functions take as input the same arguments as function d2(), adding the cash amount $Q = Q$, and output the option price. Function callCON() is

```
Function callCON(s As Double, sigma As Double, r As Double, K As Double, _
                Q As Double, tau As Double) As Double
    Dim d As Double, Nd As Double
    d = d2(s, sigma, r, K, tau)
    Nd = Application.NormSDist(d) ' std normal CDF
    callCON = Q * Exp(-r * tau) * Nd
End Function
```

The standard normal CDF is computed on the line having as comment “std normal CDF” by using the worksheet function NormSDist(). The CON put pricing function putCON() is obtained by CON put-call parity.

```
Function putCON(s As Double, sigma As Double, r As Double, K As Double, _
                Q As Double, tau As Double) As Double
    Dim C As Double
    C = callCON(s, sigma, r, K, Q, tau)
    putCON = -C + Q * Exp(-r * tau)
End Function
```

Digital AON options

As in the case of CON options, the implementation in of the pricing formulae of theorem 6.10 and corollary 6.11 is straightforward. The AON call pricing function callAON() and the corresponding put pricing function putAON(), obtained by AON put-call parity, take as input the same arguments as the corresponding CON pricing functions, with the exception of $Q = Q$, which makes no sense for AON options, and output the two prices $\text{callAON} = C_{\text{AON}}$ and $\text{putAON} = P_{\text{AON}}$.

```
Function callAON(s As Double, sigma As Double, r As Double, K As Double, _
                tau As Double) As Double
    Dim d As Double, Nd As Double
    d = d1(s, sigma, r, K, tau)
    Nd = Application.NormSDist(d)
    callAON = s * Nd
End Function

Function putAON(s As Double, sigma As Double, r As Double, K As Double, _
                tau As Double) As Double
    Dim C As Double
    C = callAON(s, sigma, r, K, tau)
    putAON = -C + s
End Function
```

Plain vanilla options

The plain vanilla call option pricing function `callPV()` is implemented using the Black and Scholes formula of theorem 6.13. Function `putPV()` computes the put price by using the put-call parity argument of theorem 6.1(iv). Inputs of the two functions are the usual ones.

```
Function callPV(s As Double, sigma As Double, r As Double, K As Double, _
               tau As Double) As Double
    Dim dd1 As Double, dd2 As Double, Nd1 As Double, Nd2 As Double
    Call d1d2(s, sigma, r, K, tau, dd1, dd2)
    Nd1 = Application.NormSDist(dd1)
    Nd2 = Application.NormSDist(dd2)
    callPV = s * Nd1 - K * Exp(-r * tau) * Nd2
End Function

Function putPV(s As Double, sigma As Double, r As Double, K As Double, _
               tau As Double) As Double
    Dim C As Double
    C = callPV(s, sigma, r, K, tau)
    putPV = C - s + K * Exp(-r * tau)
End Function
```

6.4.3 Greeks

Next we implement the Greeks.

Forward contract

We start with the Greeks of the long forward contract and implement theorem 6.17. As the first three Greeks are constant, functions `deltaFwd()` = Δ_F , `gammaFwd()` = Γ_F , and `vegaFwd()` = $\mathcal{V}_F$ have no input arguments. For a similar reason, the remaining two functions `thetaFwd()` = Θ_F and `rhoFwd()` = ρ_F take only $r = r$, $K = K$, and $\tau = \tau$.

```
Function deltaFwd() As Double
    deltaFwd = 1#
End Function

Function gammaFwd() As Double
    gammaFwd = 0#
End Function

Function vegaFwd() As Double
    vegaFwd = 0#
End Function

Function thetaFwd(r As Double, K As Double, tau As Double) As Double
    thetaFwd = -r * K * Exp(-r * tau)
End Function

Function rhoFwd(r As Double, K As Double, tau As Double) As Double
    rhoFwd = tau * K * Exp(-r * tau)
End Function
```

Plain vanilla options

The code for the Greeks of the plain vanilla call option is a straightforward implementation of the formulae of theorem 6.21. Input arguments are the same as for the pricing functions. Function N' is computed by using the Excel worksheet function `Norm_S_Dist()` (at lines with

comment “std normal PDF”); it takes two argument: the first one is where N' is computed, the second argument is a Boolean and if True the function returns the standard normal CDF, if False, the corresponding PDF.

```
Function deltaPVcall(s As Double, sigma As Double, r As Double, _
                    K As Double, tau As Double) As Double
    Dim d As Double
    d = d1(s, sigma, r, K, tau)
    deltaPVcall = Application.NormSDist(d)
End Function
```

```
Function gammaPVcall(s As Double, sigma As Double, r As Double, _
                    K As Double, tau As Double) As Double
    Dim d As Double, Nprimed1 As Double
    d = d1(s, sigma, r, K, tau)
    Nprimed1 = Application.Norm_S_Dist(d, False) ' std normal PDF
    gammaPVcall = Nprimed1 / (s * sigma * Sqr(tau))
End Function
```

```
Function vegaPVcall(s As Double, sigma As Double, r As Double, _
                    K As Double, tau As Double) As Double
    Dim d As Double, Nprimed1 As Double
    d = d1(s, sigma, r, K, tau)
    Nprimed1 = Application.Norm_S_Dist(d, False) ' std normal PDF
    vegaPVcall = Sqr(tau) * s * Nprimed1
End Function
```

```
Function thetaPVcall(s As Double, sigma As Double, r As Double, _
                    K As Double, tau As Double) As Double
    Dim dd1 As Double, dd2 As Double, Nprimed1 As Double, Nd2 As Double, _
        theta1 As Double, theta2 As Double
    Call d1d2(s, sigma, r, K, tau, dd1, dd2)
    Nprimed1 = Application.Norm_S_Dist(dd1, False) ' std normal PDF
    Nd2 = Application.NormSDist(dd2)
    theta1 = -sigma * s * Nprimed1 / (2 * Sqr(tau))
    theta2 = -r * K * Exp(-r * tau) * Nd2
    thetaPVcall = theta1 + theta2
End Function
```

```
Function rhoPVcall(s As Double, sigma As Double, r As Double, _
                    K As Double, tau As Double) As Double
    Dim d As Double, Nd2 As Double
    d = d2(s, sigma, r, K, tau)
    Nd2 = Application.NormSDist(d)
    rhoPVcall = tau * K * Exp(-r * tau) * Nd2
End Function
```

Greeks for plain vanilla put options are obtained by put-call parity.

```
Function deltaPVput(s As Double, sigma As Double, r As Double, _
                    K As Double, tau As Double) As Double
    deltaPVput = deltaPVcall(s, sigma, r, K, tau) - deltaFwd()
End Function
```

```
Function gammaPVput(s As Double, sigma As Double, r As Double, _
                    K As Double, tau As Double) As Double
    gammaPVput = gammaPVcall(s, sigma, r, K, tau) - gammaFwd()
End Function
```

```
Function vegaPVput(s As Double, sigma As Double, r As Double, _
```

```

        K As Double, tau As Double) As Double
    vegaPVput = vegaPVcall(s, sigma, r, K, tau) - vegaFwd()
End Function

Function thetaPVput(s As Double, sigma As Double, r As Double, _
    K As Double, tau As Double) As Double
    thetaPVput = thetaPVcall(s, sigma, r, K, tau) - thetaFwd(r, K, tau)
End Function

Function rhoPVput(s As Double, sigma As Double, r As Double, _
    K As Double, tau As Double) As Double
    rhoPVput = rhoPVcall(s, sigma, r, K, tau) - rhoFwd(r, K, tau)
End Function

```

Digital CON options

Implementation of CON call Greeks is done using explicit formulae of theorem 6.25.

```

Function deltaCONcall(s As Double, sigma As Double, r As Double, _
    K As Double, Q As Double, tau As Double) As Double
    Dim d As Double, Nprimed2 As Double
    d = d2(s, sigma, r, K, tau)
    Nprimed2 = Application.Norm_S_Dist(d, False)
    deltaCONcall = Q * Exp(-r * tau) * Nprimed2 / (s * sigma * Sqr(tau))
End Function

```

```

Function gammaCONcall(s As Double, sigma As Double, r As Double, _
    K As Double, Q As Double, tau As Double) As Double
    Dim dd1 As Double, dd2 As Double, Nprimed2 As Double, _
        num As Double, den As Double
    Call d1d2(s, sigma, r, K, tau, dd1, dd2)
    Nprimed2 = Application.Norm_S_Dist(dd2, False)
    num = -dd1 * Q * Exp(-r * tau) * Nprimed2
    den = s ^ 2 * sigma ^ 2 * tau
    gammaCONcall = num / den
End Function

```

```

Function vegaCONcall(s As Double, sigma As Double, r As Double, _
    K As Double, Q As Double, tau As Double) As Double
    Dim dd1 As Double, dd2 As Double, Nprimed2 As Double
    Call d1d2(s, sigma, r, K, tau, dd1, dd2)
    Nprimed2 = Application.Norm_S_Dist(dd2, False)
    vegaCONcall = -dd1 * Q * Exp(-r * tau) * Nprimed2 / sigma
End Function

```

```

Function thetaCONcall(s As Double, sigma As Double, r As Double, _
    K As Double, Q As Double, tau As Double) As Double
    Dim dd1 As Double, dd2 As Double, Nprimed2 As Double, Nd2 As Double, _
        Qdisc As Double, num As Double, den As Double
    Call d1d2(s, sigma, r, K, tau, dd1, dd2)
    Nprimed2 = Application.Norm_S_Dist(dd2, False)
    Nd2 = Application.NormSDist(dd2)
    Qdisc = Q * Exp(-r * tau)
    num = (dd1 * sigma * Sqr(tau) - 2 * r * tau) * Nprimed2
    den = 2 * sigma * tau * Sqr(tau)
    thetaCONcall = Qdisc * (num / den + r * Nd2)
End Function

```

```

Function rhoCONcall(s As Double, sigma As Double, r As Double, _

```

```

        K As Double, Q As Double, tau As Double) As Double
Dim d As Double, Nprimed2 As Double, Nd2 As Double, Qdisc As Double
d = d2(s, sigma, r, K, tau)
Nprimed2 = Application.Norm_S_Dist(d, False)
Nd2 = Application.NormSDist(d)
Qdisc = Q * Exp(-r * tau)
rhoCONcall = Qdisc * (Sqr(tau) * Nprimed2 / sigma - tau * Nd2)
End Function

```

For put options, we use the formulae of corollary 6.26, coming from CON put-call parity.

```

Function deltaCONput(s As Double, sigma As Double, r As Double, _
        K As Double, Q As Double, tau As Double) As Double
    deltaCONput = -deltaCONcall(s, sigma, r, K, Q, tau)
End Function

```

```

Function gammaCONput(s As Double, sigma As Double, r As Double, _
        K As Double, Q As Double, tau As Double) As Double
    gammaCONput = -gammaCONcall(s, sigma, r, K, Q, tau)
End Function

```

```

Function vegaCONput(s As Double, sigma As Double, r As Double, _
        K As Double, Q As Double, tau As Double) As Double
    vegaCONput = -vegaCONcall(s, sigma, r, K, Q, tau)
End Function

```

```

Function thetaCONput(s As Double, sigma As Double, r As Double, _
        K As Double, Q As Double, tau As Double) As Double
    thetaCONput = -thetaCONcall(s, sigma, r, K, Q, tau) + _
        r * Q * Exp(-r * tau)
End Function

```

```

Function rhoCONput(s As Double, sigma As Double, r As Double, _
        K As Double, Q As Double, tau As Double) As Double
    rhoCONput = -rhoCONcall(s, sigma, r, K, Q, tau) - tau * Q * Exp(-r * tau)
End Function

```

Digital AON options

Implementation of AON Greeks for calls is done using explicit formulae of theorem 6.27.

```

Function deltaAONcall(s As Double, sigma As Double, r As Double, _
        K As Double, tau As Double) As Double
Dim d As Double, Nd1 As Double, Nprimed1 As Double
d = d1(s, sigma, r, K, tau)
Nd1 = Application.NormSDist(d)
Nprimed1 = Application.Norm_S_Dist(d, False)
deltaAONcall = Nd1 + Nprimed1 / (sigma * Sqr(tau))
End Function

```

```

Function gammaAONcall(s As Double, sigma As Double, r As Double, _
        K As Double, tau As Double) As Double
Dim dd1 As Double, dd2 As Double, Nprimed1 As Double
Call d1d2(s, sigma, r, K, tau, dd1, dd2)
Nprimed1 = Application.Norm_S_Dist(dd1, False)
gammaAONcall = -dd2 * Nprimed1 / (s * sigma ^ 2 * tau)
End Function

```

```

Function vegaAONcall(s As Double, sigma As Double, r As Double, _
        K As Double, tau As Double) As Double

```

```

    Dim dd1 As Double, dd2 As Double, Nprimed1 As Double
    Call d1d2(s, sigma, r, K, tau, dd1, dd2)
    Nprimed1 = Application.Norm_S_Dist(dd1, False)
    vegaAONcall = -s * dd2 * Nprimed1 / sigma
End Function

Function thetaAONcall(s As Double, sigma As Double, r As Double, _
    K As Double, tau As Double) As Double
    Dim dd1 As Double, dd2 As Double, Nprimed1 As Double, _
        num As Double, den As Double
    Call d1d2(s, sigma, r, K, tau, dd1, dd2)
    Nprimed1 = Application.Norm_S_Dist(dd1, False)
    num = s * (dd2 * sigma * Sqr(tau) - 2 * r * tau) * Nprimed1
    den = 2 * sigma * tau * Sqr(tau)
    thetaAONcall = num / den
End Function

Function rhoAONcall(s As Double, sigma As Double, r As Double, _
    K As Double, tau As Double) As Double
    Dim d As Double, Nprimed1 As Double
    d = d1(s, sigma, r, K, tau)
    Nprimed1 = Application.Norm_S_Dist(d, False)
    rhoAONcall = s * Sqr(tau) * Nprimed1 / sigma
End Function

Put option Greeks are obtained from AON put-call parity (corollary 6.28)

Function deltaAONput(s As Double, sigma As Double, r As Double, _
    K As Double, tau As Double) As Double
    deltaAONput = 1# - deltaAONcall(s, sigma, r, K, tau)
End Function

Function gammaAONput(s As Double, sigma As Double, r As Double, _
    K As Double, tau As Double) As Double
    gammaAONput = -gammaAONcall(s, sigma, r, K, tau)
End Function

Function vegaAONput(s As Double, sigma As Double, r As Double, _
    K As Double, tau As Double) As Double
    vegaAONput = -vegaAONcall(s, sigma, r, K, tau)
End Function

Function thetaAONput(s As Double, sigma As Double, r As Double, _
    K As Double, tau As Double) As Double
    thetaAONput = -thetaAONcall(s, sigma, r, K, tau)
End Function

Function rhoAONput(s As Double, sigma As Double, r As Double, _
    K As Double, tau As Double) As Double
    rhoAONput = -rhoAONcall(s, sigma, r, K, tau)
End Function

```

6.5 Exercises

1. Prove the put-call parity relation for digital AON options (theorem 6.1(iii)), using a similar argument as the one used for the proof of the put-call parity relation for digital CON options.
2. Prove the replication of a plain vanilla put as a portfolio of digital put options (theorem 6.1(vi)), using as similar argument as the one used for the proof of the replication of a plain vanilla call as a portfolio of digital call options.

3. Prove the replication of a plain vanilla put as a portfolio of digital put options (theorem 6.1(vi)), using the parity relations for digital CON, digital AON, and plain vanilla options.
4. Consider the T -claim with contract function $\Phi(s) = \max(s, K)$; it defines a contract with long position in the underlying and with a minimum guarantee K . Replicate the contract with a long position in ZCBs and a plain vanilla call option. How do you interpret this result?
5. Consider the T -claim of exercise 4, and replicate it with a buy and hold portfolio composed by a long position on the underlying and a plain vanilla put option. How do you interpret this result?
6. Consider the T -claim of exercise 4, and replicate it with digital options.
- * 7. Prove that any European contingent claim with continuous and piecewise linear payoff function can be replicated by a portfolio of plain vanilla options.
- * 8. Prove that any European contingent claim with piecewise linear payoff function can be replicated by a portfolio of digital options.
9. Consider theorem 6.4, modify the hypothesis from ATMF to ATM and assume positive interest rates. How should the items (i) and (ii) of the thesis be changed?
10. Repeat exercise 9 without the assumption of positive interest rates.
11. Assume the Black & Scholes model, consider the T -claim of exercise 4 and price it using the result of that exercise and theorem 6.13.
12. Assume the Black & Scholes model, consider the T -claim of exercise 4 and price it using the result of exercise 5 and theorem 6.14.
13. Prove theorem 6.16.
14. Prove theorem 6.17.
15. Prove lemma 6.19 by starting from equation $d_2 = d_1 - \sigma\sqrt{\tau}$.
16. Prove lemma 6.20.
17. In figure 6.3(c) the three graphs of $\Gamma(s)$ have all a global maximum. This property holds in general; find this maximum and the level of s for which it occurs.
18. In figure 6.3(d) the three graphs of $\mathcal{V}(s)$ have all the same global maximum at $\mathcal{V} = K e^{-r\tau} \sqrt{\frac{\tau}{2\pi}}$. This property holds in general; prove it and find the level of s for which it occurs.
19. In figure 6.3(e) the three graphs of $\Theta(s)$ have all a global minimum. This property holds in general; find this minimum and the level of s for which it occurs.
20. In figure 6.3(e) the three graphs of $\Theta(s)$ have all the same horizontal asymptote at $\Theta = -rK e^{-r\tau}$ for $s \rightarrow +\infty$. This property holds in general; prove it.
21. In figure 6.3(f) the three graphs of $\rho(s)$ have all the same horizontal asymptote at $\rho = \tau K e^{-r\tau}$ for $s \rightarrow +\infty$. This property holds in general; prove it.
22. Work out the details in the proof of corollary 6.22.
23. Prove corollary 6.22 by using theorem 6.1(vi) and the pricing formulae for digital put options.
24. Prove the formula for ρ_C in theorem 6.21, by emulating the proof of the formula for Δ_C .
25. Assume the Black & Scholes model, consider the T -claim of exercise 4 and compute its Greeks using the result of that exercise and theorem 6.21.

- 26.** Assume the Black & Scholes model, consider the T -claim of exercise 4 and compute its Greeks using the result of exercise 5 and corollary 6.22.
- 27.** Prove theorem 6.25. To simplify calculations, use where needed lemma 6.19, lemma 6.20, and notice that $N''(x) = -x N'(x)$, for every $x \in \mathbb{R}$.
- 28.** Prove theorem 6.27. Start from the parity relation 6.1(v) and use the formula for the Greeks of plain vanilla options and CON options. To simplify calculations, use where needed the Zero lemma, lemma 6.19, and lemma 6.20.